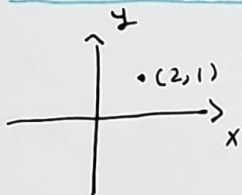


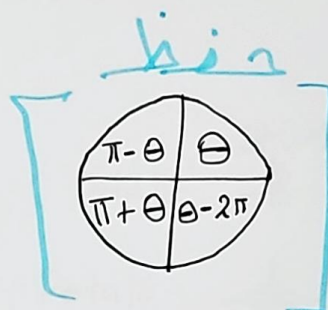
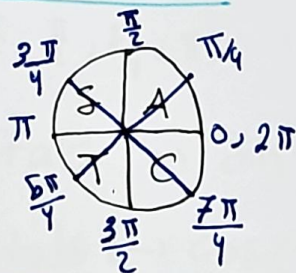
① [Polar coordinates]

[10.3] [الأحداث القطبية]

$P(x, y)$
Cartesian Coord

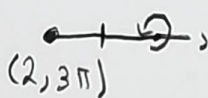


$P(r, \theta)$
Polar Coord

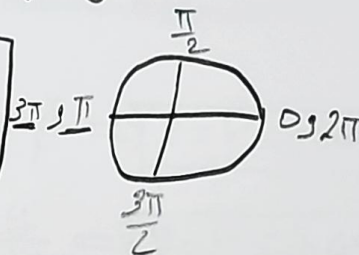


(1) - Plot the point whose Polar coordinates are $(2, 3\pi)$

[Ex]

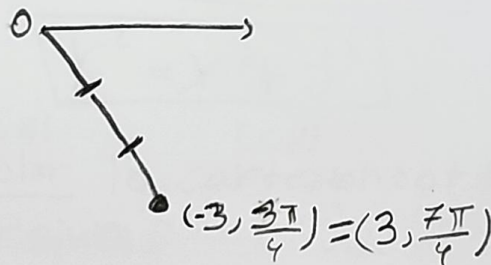


إذا لمي نرسم نقطة
نحو θ بعد ذلك نقط
النقطة نفس السؤال



$$(2) \cdot (-3, \frac{3\pi}{4}) = (3, \frac{3\pi}{4} + \pi) = (3, \frac{7\pi}{4})$$

إذا كانت (r) سالبة نقولها
+ π أو موجب ونزيد
بجدها نرسم. 😊 😊



Def: A Polar curve $r = f(\theta)$ is **symmetric**
about the **polar axis**.

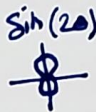
IF $f(-\theta) = f(\theta)$

(2)

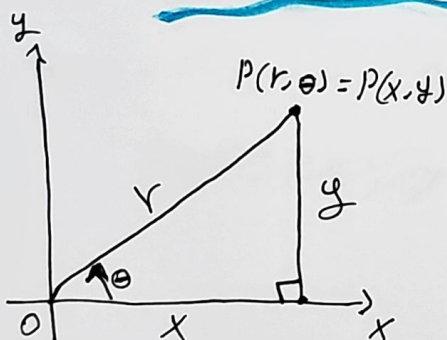
[Note]: عندما تكون الزاوية θ نتحرك بمقدار قيمة $(0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi)$

* عندما تكون الزاوية 2θ نتحرك بمقدار قيمة $(0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \dots)$

مثال

* $\sin(n\theta) = n \text{ Petals} \rightarrow \sin(2\theta) = 2 \text{ Petals}$ 

* $\cos(n\theta) = 2n \text{ Petals} \rightarrow \cos(2\theta) = 4 \text{ Petals}$ 



$$\cos(\theta) = \frac{x}{r} \Rightarrow \boxed{x = r \cos(\theta)}$$

$$\sin(\theta) = \frac{y}{r} \Rightarrow \boxed{y = r \sin(\theta)}$$

$$\boxed{\tan(\theta) = \frac{y}{x}}$$

$$\boxed{r^2 = x^2 + y^2}$$

To convert from $\overset{(r, \theta)}{\text{Polar}}$ To $\overset{(x, y)}{\text{Cartesian}}$: or to find $(x \text{ and } y)$
 $x = r \cos(\theta) / y = r \sin(\theta)$

To convert from Cartesian to Polar: or to find $(r \text{ and } \theta)$
 $r^2 = x^2 + y^2 / \tan(\theta) = \frac{y}{x}$

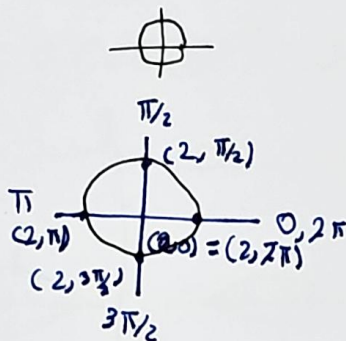
④ [1 ج 1 د]

[[Circles:

- $r = a, a > 0$

Ex: $r = 2$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	2	2	2	2	2



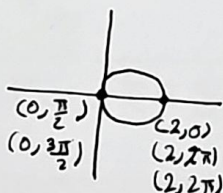
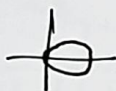
- $r = \pm a \cos(\theta)$

$r = a \cos(\theta)$

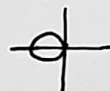
Ex: $r = 2 \cos(\theta)$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	2	0	-2	0	2

$\pi + \pi$
 2π



$r = -a \cos(\theta)$



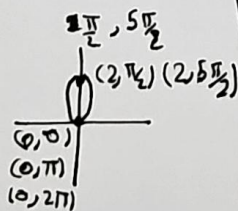
- $r = \pm a \sin(\theta)$

$r = a \sin(\theta)$

Ex: $r = 2 \sin(\theta)$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	0	2	0	-2	0

$3\pi/2 + \pi$
 $= \frac{5\pi}{2}$



$r = -a \sin(\theta)$



↑ $u \rightarrow \infty$

[2]-Cardio/s: (Heart shape ♥)

- $r = a \pm a \cos(\theta) \stackrel{\text{or}}{=} a [1 \pm \cos(\theta)]$

$$r = a + a \cos(\theta)$$

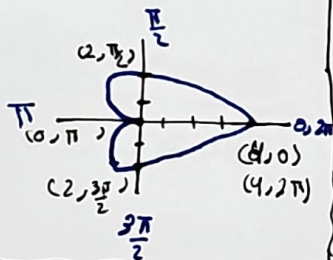
Ex: $2 + 2\cos(\theta)$



$$r = a - a \cos(\theta)$$

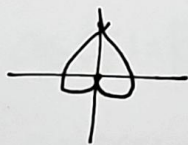


$\ominus$	0	$\pi/2$	π	$3\pi/2$	2π
r	4	2	0	2	4



- $r = a \pm a \sin(\theta) \text{ or } a[\frac{1}{2} \pm \cos(\theta)]$

$$r = a + a \sin(\theta)$$

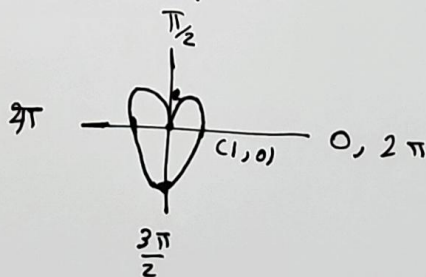


$$r = a \bar{a} \sin(\theta)$$

Ex: $1 - \sin(\theta)$



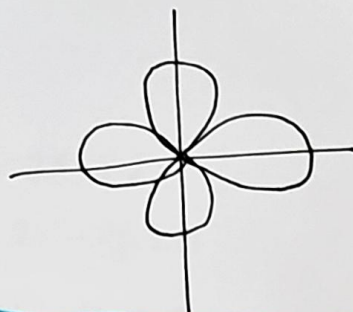
θ	0	$\pi/2$	π	$3\pi/2$	2π
r	1	0	1	2	1



⑥

[3]-Rose : [Petal 8]

• $r = \cos(2\theta)$



[Tangent or Slope of Polar]

$[r = f(\theta)]$ we know $\left[y' = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \right]$ and $\begin{cases} x = r \cos(\theta) \\ y = r \sin(\theta) \end{cases}$

$\therefore y' = \frac{\frac{dr}{d\theta} \sin\theta + r \cos(\theta)}{\frac{dr}{d\theta} \cos(\theta) - r \sin(\theta)}$

To find the slope $[m]$

$\begin{cases} \frac{dx}{d\theta} = \cos(\theta) - r \sin(\theta) \\ \frac{dy}{d\theta} = \sin\theta + r \cos(\theta) \end{cases}$

1) Find an eq. of T.L to the curve $r = 1 + \sin(\theta)$, $\theta = \pi/2$

[Ex]

$r = f(\theta) = 1 + \sin(\theta) \longrightarrow r(\pi/2) = 1 + \sin(\pi/2) = 1 + 1 = 2$

$\frac{dr}{d\theta} = \cos(\theta) \Big|_{\theta=\pi/2} = \cos(\pi/2) = 0$

$x = r \cos(\theta) \Big|_{\theta=\pi/2} = 2 \cos(\pi/2) = 0$

$y = r \sin(\theta) \Big|_{\theta=\pi/2} = 2 \sin(\pi/2) = 2$

$\therefore y' = \frac{0 \sin(\pi/2) + 2 \cos(\pi/2)}{0 \cos(\pi/2) - 2 \sin(\pi/2)} = \frac{0}{-2} = 0$
m

$\therefore y - y_0 = m(x - x_0)$

$y - 2 = 0(x - 0)$

$y - 2 = 0$

$y = 2$

(I) [Areas and lengths in polar]

[10, 4]

Def: $a \leq \theta \leq b, r = f(\theta)$

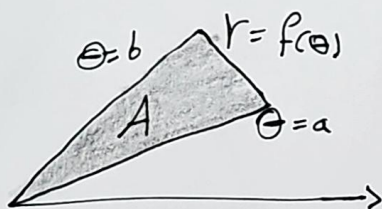
$$* \left[A = \frac{1}{2} \int_a^b [f(\theta)]^2 d\theta = \frac{1}{2} \int_a^b r^2 d\theta \right]$$

$$* \left[A = \frac{1}{2} \int_a^b (f(\theta)^2 - (g(\theta))^2) d\theta \right]$$

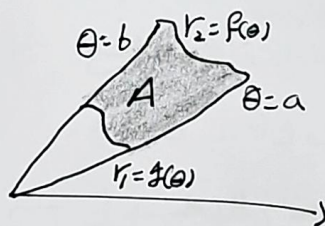
$$= \frac{1}{2} \int_a^b (r_2^2 - r_1^2) d\theta$$

* Arc length: $\left[L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \right]$

(1) - Function



(2) - Functions



$$[r_2 \geq r_1]$$

ملامحة

1- معرفة نقاط التقاطع $r=1$

(في حالة منحنى واحد $r=0$)

2- r_2 - السهم الأول
[r_1 - السهم الأخير] تأخذ سهم من نقطة الأصل ~~في اتجاه~~ باتجاه
المساحة المحددة (المطلوب) .

① [Parametric equations]

[10.1/10.2]

Def: Let f and g be cont. functions on $[a, b]$.

The equations $x = f(t)$, $y = g(t)$, $t \in [a, b]$ are called
[Parametric equations]

$$\begin{bmatrix} P(x, y) \\ = (f(t), g(t)) \end{bmatrix}$$

C consist of all point $\underline{P(t) = (f(t), g(t))}$
initial point and terminal point

$$\underline{I.P = P(a) = (f(a), g(a))}$$

[Note]

$$\underline{T.P = P(b) = (f(b), g(b))}$$

If $P(a) = P(b)$
we call the curve

C [closed curve]

(3)

[Summary]

[PL3
11.2][Let $C: x=g(t), y=h(t), a \leq t \leq b$]

[1]- $y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

[2]- [H.T.L: $y' = 0 \Rightarrow \frac{dy}{dt} = 0$
 $\therefore \frac{dy}{dt} = 0, \frac{dx}{dt} \neq 0$]

[3]- [V.T.L: $\frac{dx}{dt} = 0, \frac{dy}{dt} \neq 0$]

[4]- [The length of curve:
 $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$]

[5]- [Surface area:
x-axis $= 2\pi \int_a^b y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$
y-axis $= 2\pi \int_a^b x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$]

[6]- $y'' = \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt} y'}{\frac{dx}{dt}}$

[7]- [Area: $A = \int_a^b y dx = \int_a^b g(t) f'(t) dt$
 $\begin{cases} y = g(t) \\ x = f(t) \end{cases}$]

④ Remark

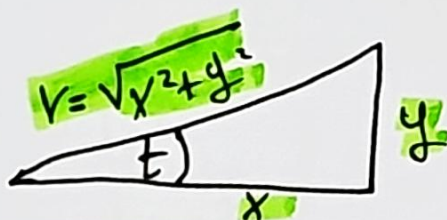
[1] $[x^2 + y^2 = a^2]$, circle with center $(0,0)$ and $[r=a]$

[or] $x = a \cos(t)$, $y = a \sin(t)$

$\left[\frac{x^2}{a^2} + \frac{y^2}{a^2} = 1\right] \rightarrow$ circle with center $(0,0)$ and $[r=a]$

[2] $\left[\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1\right] \rightarrow$ ellipse

[3]



$$\cos t = \frac{x}{r}$$

$$\sin t = \frac{y}{r}$$

$$x = r \cos(t)$$

$$y = r \sin(t)$$

(6) - Let C given by : $x = \cos t + \sin(2t)$ و $y = \sin(t) + \cos(2t)$

Find eq. of T.L at $t = \pi$

$$y - y_0 = m(x - x_0)$$

$$x_0 = x(\pi) = \cos(\pi) + \sin(2\pi) = -1 + 0 = -1$$

$$y_0 = y(\pi) = \sin(\pi) + \cos(2\pi) = 0 + 1 = 1$$

$$\frac{dx}{dt} = -\sin(t) + 2\cos(2t) \Big|_{t=\pi} = -1$$

$$\frac{dy}{dt} = \cos(t) - 2\sin(2t) \Big|_{t=\pi} = 2$$

$$m = y' = \frac{dy/dt}{dx/dt} = \frac{\cos(t) - 2\sin(2t)}{-\sin(t) + 2\cos(2t)} \Big|_{t=\pi} = -\frac{1}{2}$$

$$y - 1 = -\frac{1}{2}(x + 1)$$

لما جردا

$$\cos(\pi) = -1$$

$$\sin(\pi) = 0$$

$$\cos(2\pi) = 1$$

$$\sin(2\pi) = 0$$

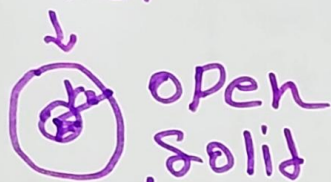
$$2\cos(2\pi) = 2$$

$$2\sin(2\pi) = 0$$

① [Volumes] [6.2]

[Volumes]

washer

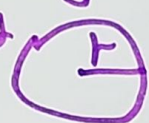


open solid

$$V_w = \pi \int_a^b (r_2^2 - r_1^2) dx \text{ (or } dy)$$

Disc

closed solid



$$V_D = \pi \int_a^b r^2 dx \text{ (or } dy)$$

Note: If the rotation [x-axis]

$$[V = \int_a^b \pi r^2 dx]$$

If the rotation [y-axis]

$$[V = \int_a^b \pi r^2 dy]$$

$$V = \pi \int_a^b (r_2)^2 - (r_1)^2 dx \text{ (or } dy)$$

ملاحظة هامة:

المسافة بين الدالة القريبة عن
محور الدوران ومحور الدوران
على حسب الأكبر يوضع أولاً
المسافة بين الدالة البعيدة عن
محور الدوران ومحور الدوران
على حسب الأكبر يوضع أولاً

2

ملاحظة عامة: اختياري للشرح يكون حسب محور الدوران

لو كان محور الدوران

$$\left[\begin{array}{l} \text{لازم نأخذ شريحة رأسية} \\ y = f(x) \text{ و } y = g(x) \\ y = 2 \end{array} \right] \delta x \leftarrow \left[\begin{array}{l} \text{or } y = c \\ x\text{-axis} \\ y = 0 \end{array} \right]$$

لو كان محور الدوران

$$\left[\begin{array}{l} \text{لازم نأخذ شريحة أفقية} \\ x = f(y) \text{ و } x = g(y) \\ x = 2 \end{array} \right] \delta y \leftarrow \left[\begin{array}{l} \text{or } x = c \\ y\text{-axis} \\ x = 0 \end{array} \right]$$

① $V = \pi \int_a^b r^2 dx \text{ (or } dy) \text{ [Disk]}$

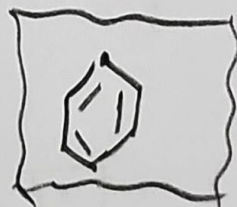
② $V = \pi \int_a^b (r_2^2 - r_1^2) dx \text{ (or } dy) \text{ [Washer]}$

r_2 - الدالة البعيدة عن محور الدوران
الدالة القريبة عن محور الدوران

r_1 - الدالة القريبة عن محور الدوران
الدالة البعيدة عن محور الدوران

[أو العكس]

لكن الكبير أولاً



(1) [Moments and centers of mass]

[8.3]

[The centroid $\rightarrow (\bar{x}, \bar{y})$]

[1] $y = f(x)$ و $y = 0$ (x-axis), $a \leq x \leq b$

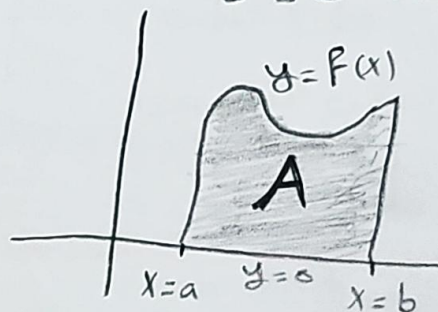
$$[A = \int_a^b f(x) dx]$$

نفس طريقة إيجاد المساحة في $cal A$

$$[\bar{x} = \frac{1}{A} \int_a^b x f(x) dx]$$

$$[\bar{y} = \frac{1}{2A} \int_a^b (f(x))^2 dx]$$

مثال على [1]

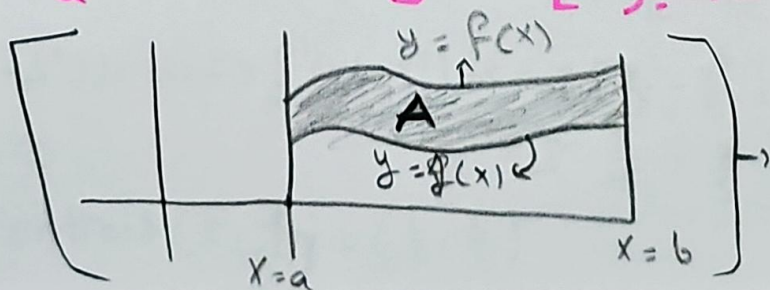


[2] $y = f(x)$ و $y = g(x)$ و $a \leq x \leq b$

$$[A = \int_a^b [f(x) - g(x)] dx] \text{ [الكبير - الصغير]}$$

$$\bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] dx \text{ [الكبير - الصغير]}$$

$$\bar{y} = \frac{1}{2A} \int_a^b [f^2 - g^2] dx \text{ [الكبير - الصغير]}$$



مثال على [2]

(2)

[Note]

IF $x = f(y)$, $x = g(y)$

$$[A = \int_a^b [f(y) - g(y)] dy]$$

$$[\bar{x} = \frac{1}{2A} \int_a^b [f^2 - g^2] dy]$$

$$[\bar{y} = \frac{1}{A} \int_a^b y [f(y) - g(y)] dy]$$

إذا كان $x = g(y)$ العكس
إذا "عكس القانونين"
[$\bar{y} \rightarrow \bar{x}$]
عكس

③

[Moments]

The origin: $M = \sum_{i=1}^n m_i x_i$

The y-axis: $M_y = \sum_{i=1}^n m_i x_i = m_1 x_1 + m_2 x_2 + \dots + m_n x_n$

The x-axis: $M_x = \sum_{i=1}^n m_i y_i = m_1 y_1 + m_2 y_2 + \dots + m_n y_n$

$$\left[\bar{x} = \frac{M_y}{m} \right]$$

$$\left[\bar{y} = \frac{M_x}{m} \right]$$

$$\left[m = \sum_{i=1}^n m_i = m_1 + m_2 + \dots + m_n \right]$$

① [Arc Length + area of Surface]

[8.1 / 8.2]

Summary

(2)

If $x = g(y)$, $c \leq y \leq d$:

[The Length]

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

[area of surface]

x-axis: $S = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

y-axis: $S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

(1)

If $y = f(x)$, $a \leq x \leq b$:

[The Length]

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

[area of surface]

x-axis: $S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

y-axis: $S = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

[Examples]

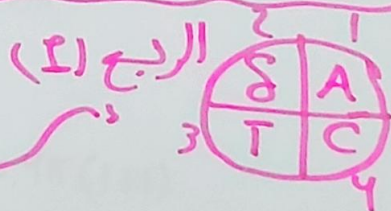
(1) - Find the arc Length of $f(x) = \ln(\sec x)$, $0 \leq x \leq \frac{\pi}{4}$

$$L = \int_0^{\pi/4} \sqrt{1 + (y')^2} dx$$

$$y' = f'(x) = \frac{1}{\sec x} \cdot \sec x \tan x = \tan x$$

$$1 + (y')^2 = 1 + \tan^2 x = \sec^2 x \implies \sqrt{1 + (y')^2} = \sqrt{\sec^2 x} = |\sec x| = \sec x$$

$$\therefore L = \int_0^{\pi/4} \sec x dx \implies = [\ln |\sec x + \tan x|]_0^{\pi/4} = \boxed{\ln |\sqrt{2} + 1|}$$



①

[Infinite sequence]

$$a_1, a_2, a_3, \dots, a_{n-1}, a_n, a_{n+1}, \dots = \{a_n\}_{n=1}^{\infty}$$

Examples: $2, 4, 6, 8, \dots, \underbrace{2n}_{a_n}, \dots = \{2n\}_{n=1}^{\infty}$ a_n
 $[n \rightarrow \infty]$
 $[a_n \rightarrow \infty]$

note:

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots = \left\{ \frac{1}{n} \right\}_{n=1}^{\infty}$$

$n \rightarrow \infty$
 $a_n = \frac{1}{n} \rightarrow 0$

$$0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots, \underbrace{\frac{n-1}{n}}_{a_n}, \dots = \left\{ \frac{n-1}{n} \right\}_{n=1}^{\infty}$$

$[n \rightarrow \infty]$
 $[a_n = \frac{n-1}{n} \rightarrow 1]$

$$1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}, \dots, (-1)^{n+1} \frac{1}{n}, \dots = \left\{ (-1)^{n+1} \frac{1}{n} \right\}_{n=1}^{\infty}$$

a_n

[لنلاحظ اننا مرة موجبه ومرة سالبة]

(2)

[factorial] تعني

$$n \rightarrow \infty$$

$$a_n = \frac{1}{n!} \rightarrow \frac{1}{\infty} = \boxed{0}$$

$$n! = n(n-1)(n-2)(n-3) \dots (3)(2)(1) \quad \leftarrow \text{بمعنى}$$

$$1! = 1 = 1$$

$$2! = (2)(1) = 2$$

$$3! = (3)(2)(1) = 6$$

$$4! = (4)(3)(2)(1) = 24$$

$$5! = (5)(4)(3)(2)(1) = 120$$

$$\Rightarrow n! = n(n-1)! = (n-1)(n-2)! \quad [\text{اختصار}]$$

(Sequence convergence) متى يكُون [تقارباً] Examples:

If $\lim_{n \rightarrow \infty} a_n = L$ (exist) we say that $\{a_n\}$ conv.

If $\lim_{n \rightarrow \infty} a_n$ fails to exist (DNE) we say that $\{a_n\}$ div.

(1) $\lim_{n \rightarrow \infty} 2n = \infty \Rightarrow \{2n\}$ is div.

(2) $\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \Rightarrow \{\frac{1}{n}\}$ is conv.

(3) $\lim_{n \rightarrow \infty} (2n-1) = \infty \Rightarrow \{2n-1\}$ is div.

(4) $\lim_{n \rightarrow \infty} \frac{n-1}{n} = \frac{\infty}{\infty} \Rightarrow f(x) = \frac{x-1}{x} \rightarrow \lim_{x \rightarrow \infty} \frac{x-1}{x} = \lim_{x \rightarrow \infty} \frac{x}{x} = 1$
 $\therefore \{\frac{n-1}{n}\}$ is conv.

(5) $\lim_{n \rightarrow \infty} 5 = 5 = 5, \therefore \{5\}$ is conv.

Theorem:

ما هي النظرية
 متى تطبق $\rightarrow$ 1- [If $\lim_{n \rightarrow \infty} |a_n| = 0$ then $\lim_{n \rightarrow \infty} a_n = 0$]
 ان اذا كان مطلق $|a_n|$ يساوي 0 اذا a_n

1- اذا كان مطلق $|a_n|$ يساوي 0 اذا a_n

يساوي 0

2- اذا كان لا يساوي 0 "ك" نستطيع تطبيق النظرية

[$\therefore \{a_n\}$ is conv. If we can use the theorem.]

③

$$\left[\lim_{n \rightarrow \infty} |a_n| = 0 \right] \rightarrow \text{proof:}$$

مع التسوية
يكون صفر

$a_n = 0$

$$-|a_n| \leq a_n \leq |a_n|$$

Show $\lim_{n \rightarrow \infty} |a_n| = 0$

$$\therefore -\lim_{n \rightarrow \infty} |a_n| = 0$$

$$\lim_{n \rightarrow \infty} a_n = 0 \quad [\text{by } \delta T]$$

Theorem: Let $\{a_n\}$ and $\{b_n\}$ be sequences of real number and let A and B be real numbers. If $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$ then,

- $\lim_{n \rightarrow \infty} (a_n \pm b_n) = A \pm B$
- $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = AB$
- $\lim_{n \rightarrow \infty} (K \cdot b_n) = KB, K$ is any real number
- $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}, B \neq 0$

يستخدم هذه النظرية في حالة وجود ليمت و 2 sequences
لكن نستطيع إيجاد الناتج ونسعى لحل

[Some Basic Limits]

[هام وحفظ]



(1) • $\lim_{h \rightarrow \infty} \frac{\ln(h)}{h} = 0$

(2) • $\lim_{h \rightarrow \infty} \sqrt[h]{h} = 1$

(3) • $\lim_{h \rightarrow \infty} (x)^{\frac{1}{h}} = 1, x > 0$

(4) • $\lim_{h \rightarrow \infty} (x)^h = \begin{cases} > 0, -1 < x < 1 \\ 1, x = 1 \\ \infty, x > 1 \\ DNE, x < -1 \end{cases}$

(5) • $\lim_{h \rightarrow \infty} \frac{(x)^h}{h!} = 0$

(6) • $\lim_{h \rightarrow \infty} \left(1 + \frac{x}{h}\right)^h = e^x$

For any $[x]$
 $\lim_{h \rightarrow \infty} \left(1 + \frac{x}{h}\right)^{yh} = e^{xy}$

(10) • $\lim_{h \rightarrow \infty} \frac{h!}{x^h} = \infty$

(7) • $\lim_{h \rightarrow \infty} \frac{\sinh\left(\frac{h}{n}\right)}{\left(\frac{h}{n}\right)} = 1$

(8) • $\lim_{h \rightarrow \infty} \frac{\tan\left(\frac{h}{n}\right)}{\left(\frac{h}{n}\right)} = 1$

(9) • $\lim_{h \rightarrow \infty} \frac{1 - \cos\left(\frac{h}{n}\right)}{\left(\frac{h}{n}\right)} = 0$

(8). $a_n = \frac{\sin^2 n}{2^n} \rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{\sin^2 n}{2^n}$
 [في نفس المسألة نعلمها] [ST]

[هام]

$$0 < \sin^2 n \leq 1$$

$$0 = \frac{0}{2^n} \leq \frac{\sin^2 n}{2^n} \leq \left[\frac{1}{2^n} \right] \text{, [By basic Limits]} = 0 \rightarrow \frac{1}{2^\infty} = \frac{1}{\infty} = 0$$

$$\therefore \lim_{n \rightarrow \infty} \frac{0}{2^n} = \lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$$

$$\therefore \lim_{n \rightarrow \infty} \frac{\sin^2 n}{2^n} = 0 \text{ by [ST]}$$

$\therefore \{a_n\}$ is conv.

(4). $a_n = \frac{n!}{n^n}$
 [هام جداً]

$$\rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n!}{n^n} = \frac{\infty}{\infty}$$

$$0 < \frac{n!}{n^n} = \frac{(n-1)(n-2)(n-3)\dots 3, 2, 1}{n \cdot n \cdot n \dots n \cdot n \cdot n}$$

[أضرب من 1]

$\frac{1}{n}$
 $\frac{1}{\infty} = 0$

$$\lim_{n \rightarrow \infty} 0 = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\therefore \text{By [ST]} \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

$\therefore \{a_n\}$ is conv.

[هام]

Note:

في هذه النوعية من المسائل لا نستطيع تطبيق
 1, 2 لأننا على سبيل المثال لا نعرف مشتقة $n!$

لذلك نستخدم:

(1) -> Approximation [تقريب]

(2) -> Squeeze (or sandwich) Theorem.

(7)

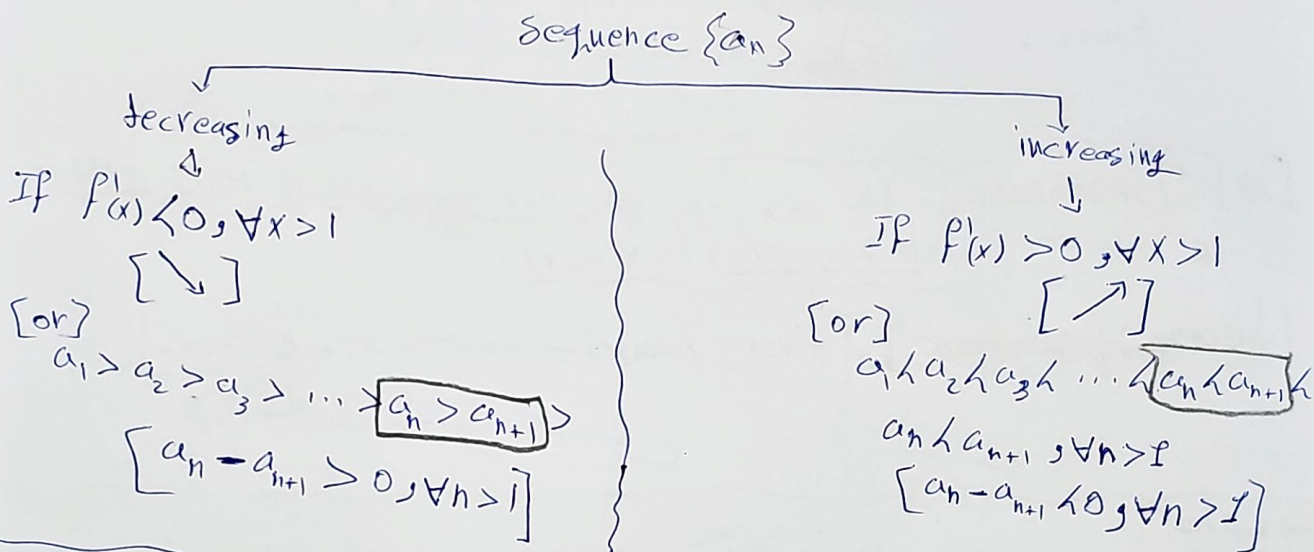
[Remark]

[سلاسل جبرية]

$$[1] \rightarrow \lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_{n-1} = [L] \rightarrow \text{we}$$

[2] - (1) A sequence $\{a_n\}$ is called increasing

(2) A sequence $\{a_n\}$ is called decreasing



[3] -> A sequence $\{a_n\}$ is bounded from above if there is a number $[M]$ such that: $[a_n \leq M \text{ for all } n]$. The number $[M]$ is called an upper bound of $\{a_n\}$.

[4] -> A sequence $\{a_n\}$ is bounded from below if there is a number $[m]$ such that: $[a_n \geq m \text{ for all } n]$. The number $[m]$ is called an lower bound of $\{a_n\}$.

[5] -> If the $\{a_n\}$ bounded from above and below, then $\{a_n\}$ is called bounded otherwise unbounded.

Lower bound $\leftarrow [m] \leq a_n \leq [M] \rightarrow$ upper bound

[Theorem]:

inc, constant

[1] $\rightarrow$ A [non-decreasing] seq of real number conv. if it is bounded from above.
إذا كانت متزايدة من أعلى و محدودة من أعلى تكون متقاربة
Seq $\nearrow$ ~~Seq~~ $\nearrow$ conv.

[2] $\rightarrow$ ^{dec, constant}

A [non-increasing] seq of real number conv. if it is bounded from below.
إذا كانت متناقص من أسفل و محدودة من أسفل تكون متقاربة
Seq $\searrow$ conv.

Def: [Monotonic]: The seq $\{a_n\}$ is called monotonic if it is either increasing or decreasing.

Theorem: [Monotonic]: [Every bounded + monotonic seq. is conv.]

والمتكافئ غير صحيح

①

[Infinite series]

Def: [Infinite series]: An infinite series is the sum of an infinite sequence of numbers. It is an expression in the form:

$$a_1 + a_2 + a_3 + \dots + a_n + \dots = \sum_{n=1}^{\infty} a_n$$

Sequences

$$a_1, a_2, a_3, \dots, a_n, \dots = \{a_n\}_{n=1}^{\infty}$$

$$\lim_{n \rightarrow \infty} a_n = L \rightarrow \{a_n\} \text{ conv.}$$

Series

$$a_1 + a_2 + a_3 + \dots + a_n + \dots = \sum_{n=1}^{\infty} a_n$$

conv. series:

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

...

$$s_{n-1} = a_1 + a_2 + a_3 + \dots + a_{n-1}$$

$$s_n = a_1 + a_2 + a_3 + \dots + a_{n-1} + a_n$$

["مجموعه"] ★

$$a_n = s_n - s_{n-1}, \quad a_1 = s_1$$

the n-th term for the seq. of Partial [Sum]

$$\lim_{n \rightarrow \infty} s_n = L \text{ (exist)}$$

$$\{s_n\} \text{ conv.} \rightarrow \sum a_n \text{ conv.}$$

$$\lim_{n \rightarrow \infty} s_n = \begin{pmatrix} \infty \\ -\infty \\ DNE \end{pmatrix}$$

$$\rightarrow \{s_n\} \text{ Div.} \rightarrow \sum a_n \text{ div. No sum.}$$

• s_n "مجموعه" اذا "مجموعه" [conv. و div.] series
 و a_n "سلسلة" اذا [conv. و div.] sequence
 و نظرياتهما.

The First test

بعض العلاقة () - () . → (1)

[يجب ان يكون احدهما $+ve$ الاول والثاني $-ve$]
 ثم نتحول الى الحالة الاولى $\rightarrow \frac{C}{C} \rightarrow (2)$

ثم تحول الى الحالة الاولى

(3)

[Geometric Series] [G.S]

number 2 the test series



ملاحظة هامة: G.S افكر بها عندما يكون عندي

(عدد)ⁿ
اونحاول نوضحها اي شكل (عدد)ⁿ
اوقسمة اي عدد على سابقة مقدار
ثابت وهو (r)Geometric series:

It is a series in the form:

$$\underbrace{a}_{=r} + \underbrace{ar}_{=r} + \underbrace{ar^2}_{=r} + ar^3 + \dots + ar^{n-1} + \dots = \sum_{n=1}^{\infty} ar^{n-1}$$

(1) [a] is the first term, $a \neq 0$

(2) [r] is the ratio

G.S

Div.

$$\left\{ \begin{array}{l} \sum_{n=1}^{\infty} ar^{n-1} \text{ is Div.} \\ \text{[IF } |r| > 1] \\ \text{[no sum]} \end{array} \right\}$$

conv.

$$\left\{ \begin{array}{l} \sum_{n=1}^{\infty} ar^{n-1} \text{ is conv.} \\ \text{[IF } |r| < 1] \\ \text{[Sum} = \frac{a}{1-r}] \end{array} \right\}$$

Examples:

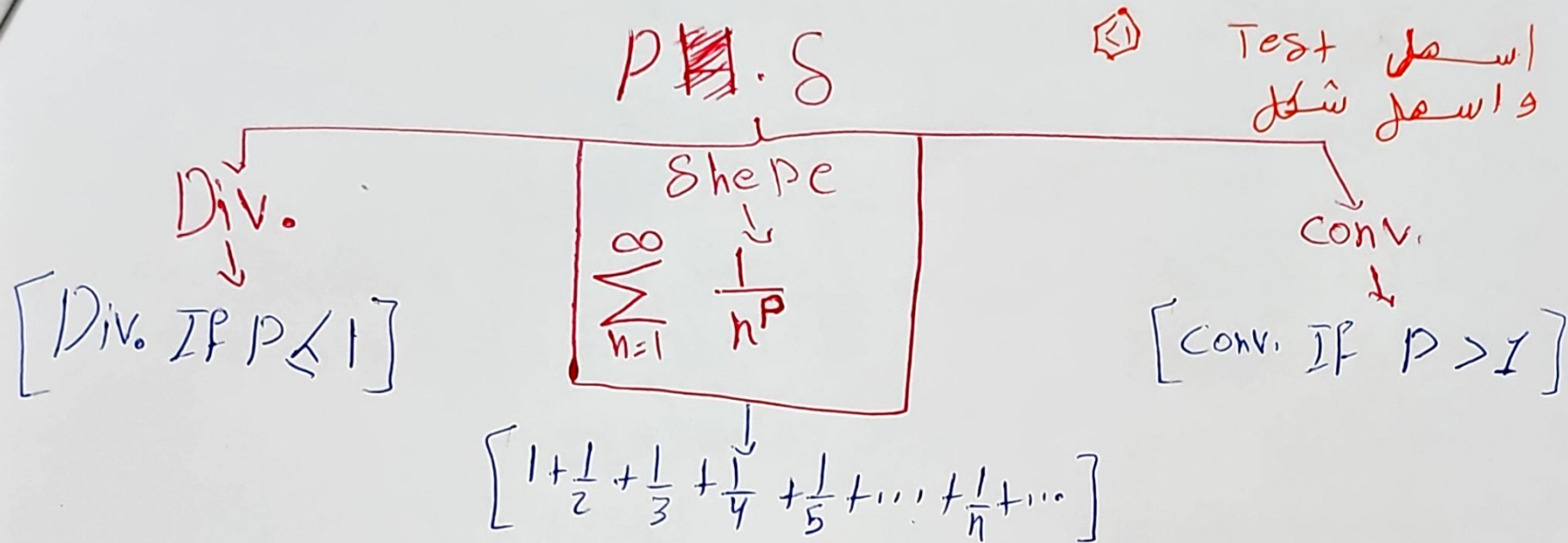
$$[1] \rightarrow \sum_{n=0}^{\infty} (-1)^n \frac{5}{4^n} \rightarrow \sum_{n=0}^{\infty} 5 \left(\frac{-1}{4} \right)^n$$

 $a = 5$ و conv. G.S

$$r = \frac{1}{4} \quad \text{because } |r| = \left| \frac{-1}{4} \right| = \frac{1}{4} < 1$$

$$\text{Sum} = \frac{a}{1-r} = \frac{5}{1 - (-\frac{1}{4})} = \frac{5}{1 + \frac{1}{4}} = \frac{5}{\frac{5}{4}} = \frac{20}{4} = \boxed{5}$$

[harmonic series] [p-series]
[Test number 3] [Test number 4]



(7)

[The n-th term test for Div.]

[Test number 4]

If $\lim_{n \rightarrow \infty} a_n \neq 0$ then $\sum_{n=1}^{\infty} a_n$ Div. — [Div. لا نستطيع لـ]

* هذا الاختبار نستعمله لـ (Div.)
* اولى في استخراجه اذا كان لدى $(\infty - \infty$ او $\frac{\infty}{\infty})$ استعمل $\frac{\infty}{\infty}$

[Examples]

[1] $\rightarrow \sum_{n=1}^{\infty} \left(\frac{n!}{(1000)^n} \right) = \frac{\infty}{\infty} \rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n!}{(1000)^n} = \infty \neq 0$ (basic limits)

$\therefore \sum a_n$ is Div. by n-th term test for div.

$\left[\lim_{n \rightarrow \infty} \frac{n!}{x^n} = \infty \right]$

[2] $\rightarrow \sum_{n=1}^{\infty} \left(\frac{2^n + 4^n}{3^n + 4^n} \right) = \frac{\infty}{\infty} \rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{2^n + 4^n}{3^n + 4^n}$

$$= \lim_{n \rightarrow \infty} \frac{\frac{2^n}{4^n} + \frac{4^n}{4^n}}{\frac{3^n}{4^n} + \frac{4^n}{4^n}} = \frac{1}{1} = 1 \neq 0$$

$\therefore \sum a_n$ is Div. by n-th term test for div.

[3] $\rightarrow \sum_{n=1}^{\infty} \left(\frac{n+1}{\sqrt{4n^2+3}} \right) = \frac{\infty}{\infty} \rightarrow \lim_{n \rightarrow \infty} \frac{n+1}{\sqrt{4n^2+3}} = \lim_{n \rightarrow \infty} \frac{(n+1)}{(n)\sqrt{4+\frac{3}{n^2}}}$

$$= \lim_{n \rightarrow \infty} \frac{\frac{n}{n} + \frac{1}{n}}{\frac{n}{n}\sqrt{4+\frac{3}{n^2}}} = \frac{1}{\sqrt{4}} = \frac{1}{2} \neq 0$$

$\therefore \sum a_n$ is div. by n-th term test for div.

[4] $\rightarrow \sum_{n=1}^{\infty} \left(\frac{n}{n-1} \right)^n = \left(\frac{\infty}{\infty} \right)^{\infty} \rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{n}{n-1} \right)^n = \lim_{n \rightarrow \infty} \left(\frac{n}{n(1-\frac{1}{n})} \right)^n$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{1-\frac{1}{n}} \right)^n = \lim_{n \rightarrow \infty} \frac{(1)^n}{\left(1-\frac{1}{n}\right)^n} = \frac{1}{e^{-1}} \neq 0$$

$\therefore \sum a_n$ is div. by n-th term test for div.

basic limits

$\left[\begin{aligned} \lim_{n \rightarrow \infty} \left(1 - \frac{x}{n}\right)^n &= e^{-x} \\ \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n &= e^x \end{aligned} \right]$

④ Theorem:

- * If $\sum a_n = A$ and $\sum b_n = B$ are convergent series.
- * $\sum a_n \pm b_n = A \pm B$ is convergent.
- * If $[k]$ is a non-zero constant then $\sum k a_n = kA$ is convergent.
- * $\text{Conv.} \pm \text{Conv.} = \text{Conv.}$
- * $\text{Div.} \pm \text{Conv.} = \text{Div.}$
- * $\text{Div.} \pm \text{Div.} = \text{may be conv. or Div.}$

Q: Give an example of two Div. Series their sum is conv.



Leibniz

①

Solve: $\sum_{n=1}^{\infty} (-1)^{2n} = 1 + 1 + 1 + \dots$
Div. G.S because

$|r| = |-1| = 1$

②

$\sum_{n=1}^{\infty} (-1)^{2n-1} = -1 + 1 + -1 + \dots$
Div. G.S because

$|r| = |-1| = 1$

$\sum_{n=1}^{\infty} (-1)^{2n} + \sum_{n=1}^{\infty} (-1)^{2n-1} = 0$

$1 + (-1) = 0$ Conv.

Example: $\sum_{n=0}^{\infty} \left(\frac{5}{2^n} - \frac{1}{3^n} \right) \rightarrow \sum_{n=0}^{\infty} \frac{5}{2^n} - \sum_{n=0}^{\infty} \frac{1}{3^n}$

$= 5 \sum_{n=0}^{\infty} \left(\frac{1}{2} \right)^n$ conv. G.S $|r| = \frac{1}{2} < 1$ $\left\{ \sum_{n=0}^{\infty} \left(\frac{1}{3} \right)^n \right.$ G.S conv. $|r| = \frac{1}{3} < 1$

Sum $= \frac{a}{1-r} = \frac{5}{1-\frac{1}{2}} = \frac{5}{\frac{1}{2}} = 10$

Sum $= \frac{a}{1-r} = \frac{1}{1-\frac{1}{3}} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$

$\therefore S = S_1 - S_2 \rightarrow 10 - \frac{3}{2} = \frac{17}{2}$

①

[Integral test]

Test number 5The integral test:

- * $f(x)$ is positive
- * $f(x)$ is cont. on $[N, \infty)$
- * $f(x)$ is decreasing for all $x \geq N$

N : the smallest integer that satisfies the condition.

[196]

* Then the series $\sum_{n=1}^{\infty} a_n$ and $\int_N^{\infty} f(x) dx$ are either both conv. or Div. or both Div.

IF $\Rightarrow$ conv. $\Rightarrow$ conv.

IF $\Rightarrow$ Div. $\Rightarrow$ Div.

ملاحظة هامة: افكر غالباً هذا الاختبار

I, E

(1) لو فرق فكامل الدالة بكل سهولة

(2) لو فرق مثلاً $u = (\sqrt{x})$ ووجد المشتقة او جزئ منها بالسؤال

①

[Alternating series]

Alternating series test: [A.S.]

Def: The alternating series is an expression of the form:

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + \dots + (-1)^{n+1} a_n + \dots$$

or

$$\sum_{n=1}^{\infty} (-1)^n a_n = -a_1 + a_2 - a_3 + a_4 + \dots + (-1)^n a_n + \dots$$

[A.S.T.]

$$\sum_{n=1}^{\infty} (-1)^{n+1} \text{ conv, IF}$$

* $\lim_{n \rightarrow \infty} a_n = 0$

* The $\{a_n\}$ is are all positive

* The a_n is are $\downarrow$ that is, $\{a_n \geq a_{n+1}\}$

$$\sum_{n=1}^{\infty} (-1)^{n+1} \text{ Div, IF}$$

لم يحقق احدي شروط conv.

[1] $\rightarrow \sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{n} \right)$ A.S.

$\hookrightarrow a_n = \frac{1}{n} \rightarrow 1 - \lim_{n \rightarrow \infty} \frac{1}{n} = 0$

2 - a_n is +ve, $\forall x \geq 1$

3 - $f(x) = \frac{1}{x}$

$f'(x) = -\frac{1}{x^2} < 0, \forall x \geq 1$

$\therefore f$ is $\downarrow \Rightarrow \{a_n\}$ is $\downarrow$

$\therefore$ ~~From~~ A.S. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$ is conv, by A.S.T.

$$(-1)^n = \begin{cases} \rightarrow 1, & n = \text{even} \\ \rightarrow -1, & n = \text{odd} \end{cases}$$

$$(-1)^{n+1} = \begin{cases} \rightarrow 1, & n = \text{odd} \\ \rightarrow -1, & n = \text{even} \end{cases}$$

Notes

Remark:

- (1) - $|(-1)^{n+1}| = |(-1)^n| = 1$
- (2) - Abs. conv means conv, and cond. conv, means conv,
- (3) - If the third condition of the previous theorem is not satisfied that is $(\lim_{n \rightarrow \infty} a_n \neq 0)$ then the series $\sum_{n=1}^{\infty} (-1)^n a_n$ is Div, by nth terms test.
- (4) - Cond. convergent means the series is conv, but not, Abs. conv.

Summary

$$\sum (-1)^n a_n$$



$$\sum |a_n| = \sum |a_n|$$

$\sum |a_n|$ Div

$\lim_{n \rightarrow \infty} a_n \neq 0$

$\therefore \sum (-1)^n a_n$
Div
by
n-th term
test for Div.

1 - $\lim_{n \rightarrow \infty} a_n = 0$

2 - $a_n > 0$

3 - a_n is +ve

$\therefore \sum (-1)^n a_n$
cond. conv,
[C.C.]

$\sum |a_n|$ conv,

$\therefore \sum (-1)^n a_n$

Abs. conv

[A.C.]

①

[Ratio and Root Test]

The Ratio test [5]: [R.T]

* Let $\sum a_n$ be a series with +ve

and suppose that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \boxed{p}$

$\sum a_n$ conv, IF $p < 1$

$\sum a_n$ Div, IF $p > 1$

[IF $p = 1$ then the test fail]

The n-th Root test [6]: [n-th R.T]

* Let $\sum a_n$ be a series with +ve

and suppose that $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \boxed{p}$

$\sum a_n$ conv, IF $p < 1$

$\sum a_n$ Div, IF $p > 1$

[IF $p = 1$ then the test fail]

[Examples]

$$[1] - \sum_{n=1}^{\infty} \frac{n!}{(10)^n}$$

$$a_{n+1} = \frac{(n+1)!}{(10)^{n+1}} = \frac{(n+1)n!}{(10)^n \cdot 10}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{(n+1)\cancel{n!}}{(10)^n \cdot 10} \cdot \frac{(10)^n \cdot \cancel{n!}}{n!} = \frac{1}{10} \lim_{n \rightarrow \infty} (n+1) = \infty > 1$$

$\therefore \sum a_n$ Div, by R.T

① [Comparison and Limit comparison test]

① Comparison test: $\underbrace{\downarrow}_{\text{Div.}} \quad \underbrace{\downarrow}_{\text{conv.}}$

$$\downarrow n \leq a_n \leq c_n \quad [\sum \downarrow n, \sum a_n, \sum c_n \text{ are +ve}]$$

If $\sum \downarrow n$ Div. then $\sum a_n$ also Div. If $\sum c_n$ conv. then $\sum a_n$ also conv.

[Note]
↓

$$\downarrow n \leq a_n \leq c_n$$

في هذا الاتجاه ممكن ان:

1- تكبير البسط

2- تكبير المقام

Div

في هذا الاتجاه ممكن ان:

1- تكبير البسط

2- تكبير المقام

conv.

[متى نستخدم هذا الاختبار]

1- $\sin(n)$ او $\cos(n)$

2- $n!$

نفسر في استخدام C.T

[Limit Comparison Test [L.C.T]]

(4) [مهم]

a_n and b_n $\rightarrow$ b_n اختبار

1- IF $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$ are both conv, or Div,

2- IF $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ conv, then $\sum a_n$ conv,

3- IF $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ Div, then $\sum a_n$ is ~~conv~~ ^{Div},

ملاحظة هامة: في L.C.T: كيف نختار b_n : ★ ★

1- ا- أكبر واقلوى اس في البسط

الكل واقلوى اس في المقام

إذا رأينا: $\cos(\frac{1}{n})$ او $\sin(\frac{1}{n})$

نفكر في استخدام L.C.T او T.C

لا نستخدم L.C.T إذا رأينا :

$n!$: لاننا لانعرف مشتقتها

$\cos(n)$ او $\sin(n)$

لانه $\cos(\infty)$ و $\sin(\infty)$ غير معروفه .

① [Taylor and Maclaurin series] [11/10]

[Taylor series]: generated by the function f at the point a is:

is: $[(n) \text{ سلسله }]$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)(x-a)^n}{n!}$$

$$= f(a) + \frac{f'(a)(x-a)^1}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^{(n)}(a)(x-a)^n}{n!} + \dots$$

[Maclaurin series]: If $a=0$, then we get the Maclaurin series that is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)x^n}{n!}$$

$$= f(0) + \frac{f'(0)x^1}{1!} + \frac{f''(0)x^2}{2!} + \dots + \frac{f^{(n)}(0)x^n}{n!} + \dots$$

Remark:

- * $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
- * $\cos(x \pm y) = \cos x \cos y \pm \sin x \sin y$
- * $1, -1, 1, -1, \dots, (-1)^{n+1}, \dots$
- * $-1, 1, -1, 1, \dots, (-1)^n, \dots$
- * $0, 2, 4, 6, \dots, 2n, \dots$
- * $1, 3, 5, 7, \dots, 2n+1, \dots$

note: هذا السكشن سنحاول فيه الـ power series

الـ power series شكل مختلف (واختلافه) المشهور

n degree Taylor Polynomial of $f(x)$ at $x=a$

$$T_n(x) = f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^{(n)}(a)(x-a)^n}{n!} + \dots$$

The binomial series: Conv, if $|x| < 1$

If k is any real number and $|x| < 1$, then

$$[(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n]$$

$$= 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3 + \dots$$

$$[\text{where } \binom{k}{n} = \frac{k(k-1)(k-2)\dots(k-n+1)}{n!}]$$

Theorem: If f has a power series representation at a , that is, if

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n, \quad |x-a| < R$$

$$c_n = \frac{f^{(n)}(a)}{n!}$$

$$\left\{ \begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \\ c_n &= \frac{f^{(n)}(a)}{n!} \end{aligned} \right\} \xrightarrow{\text{Taylor Series}} \Rightarrow \boxed{f^{(n)}(a) = c_n n!}$$

Find the sum of each series:

(a) - $\sum_{n=0}^{\infty} \binom{3/2}{n} \frac{7^n}{3^{2n}}$

(b) - $\sum_{n=0}^{\infty} (-1)^n \frac{\pi^{2n+1}}{4^{2n+1} (2n+1)!}$

(a) - Using the Binomial series $k = 3/2$, $x = \frac{7}{9} < 1 \therefore \text{conv.}$

$$\begin{aligned} \sum_{n=0}^{\infty} \binom{3/2}{n} \frac{7^n}{3^{2n}} &= \sum_{n=0}^{\infty} \binom{3/2}{n} \frac{7^n}{9^n} = \sum_{n=0}^{\infty} \binom{3/2}{n} \left(\frac{7}{9}\right)^n = (1+x)^k \\ &= \left(1 + \frac{7}{9}\right)^{3/2} \\ &= \left(\frac{16}{9}\right)^{3/2} = \frac{64}{27} \end{aligned}$$

(b) - $\sum_{n=0}^{\infty} (-1)^n \frac{\pi^{2n+1}}{4^{2n+1} (2n+1)!}$ using Maclaurin series [Power series (4)]

$$= \left[\sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{\pi}{4}\right)^{2n+1}}{(2n+1)!} \right] = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$\sin(u) = \sum_{n=0}^{\infty} \frac{(-1)^n u^{2n+1}}{(2n+1)!}$$

If $f(x) = \sum_{n=0}^{\infty} \frac{(x-2)^n}{n^3}$, then $f^{(10)}(2) = ?$

$$c_n = \frac{f^{(n)}(a)}{n!}$$

$$c_n = \frac{1}{n^3}, \quad n \leq 10, \quad a = 2$$

$$\begin{aligned} f^{(10)}(2) &= c_n \cdot n! \Rightarrow = c_{10} \cdot (10)! \\ &= \frac{1}{(10)^3} \cdot (10)! = \frac{1}{10^3} \cdot 1! \\ &= \frac{1}{100} \end{aligned}$$

①

[The power series]

Power series
متسلسلات القوى

$$\sum_{n=0}^{\infty} a_n (x-c)^n \quad \text{or} \quad \sum_{n=0}^{\infty} a_n x^n, \quad \boxed{x \neq 0}$$

لاحظ: عاملها كالتالي $\rightarrow x$ We fixed x
لأن لو $x=0$ راح تكون conv, series

[كيفنا ايجاد]

②

Interval of conv,

①

Radius of conv,

① لا يجاوم ① نستخدم فقط

يعني نفوف x بالقيمة بالسيريس وندرسها باه نوع من انواع الاختبار السابقة.

$$\boxed{\begin{matrix} x = -R \\ x = R \end{matrix}}$$

③

Ratio test

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

n-th R.T

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

② سوف تظهر لنا $-R < x < R$

③ لازم ندرس الا x عند الامتداد

يمكن مختلفة
يمكن مختلفة
 $I = (a, b)$

④ - اذا كانت كلاهما Div او احدهما نفتح الفترة ونخلقها

واذا كانت كلاهما conv, او احدهما لا يجاوم, interval of conv,

$$\boxed{R = \frac{b-a}{2}}$$

⑤ لا يجاوم Radius = R

Power series $\sum a_n (x-c)^n$ is conv,

$$\text{If } |x-c| < R$$

$$-R < x-c < R$$

$$\boxed{c-R < x < c+R}$$

[نظام]

②

[Summary]

$\sum a_n(x-a)^n$ or $\sum a_n x^n$

I

$x \in [a, b]$
or
 (a, b)
or
 $[a, b)$
or
 $(a, b]$

③

I
 $x = a$
نقطة

إذا كانت النهاية
 $\infty =$

$R = 0$
 $I = x = a$

ننظر عن القيمة الق

تخالي السيريس مفر

ال x مفر وتكون

عندها conv,

②

I ①
 $x \in (-\infty, \infty)$

إذا كانت

النهاية $= 0$

$I = R$
 $= (-\infty, \infty)$

$R = \infty$

لازم تكون قيمة

النهاية تساوي I

$$R = \frac{b-a}{2}$$

ونحل المتباينة

وتكون مغلقة إذا كانت conv,

ومفتوحة إذا كانت Div,

1

Representation of functions as power series in power of x

[II, 2]

① - $\frac{1}{1+u} = \sum_{n=0}^{\infty} (-1)^n u^n \longrightarrow |u| < 1 \longrightarrow I = (-1, 1) \longrightarrow R = 1$

② - $\frac{1}{1-u} = \sum_{n=0}^{\infty} u^n \longrightarrow |u| < 1 \longrightarrow I = (-1, 1) \longrightarrow R = 1$

③ - $\ln(1+u) = \sum_{n=0}^{\infty} \frac{(-1)^n u^{n+1}}{n+1} \longrightarrow I = (-1, 1] \longrightarrow R = 1$

④ - $\tan^{-1} u = \sum_{n=0}^{\infty} \frac{(-1)^n u^{2n+1}}{2n+1} \longrightarrow I = [-1, 1] \longrightarrow R = 1$

⑤ - $e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!} \longrightarrow I = \mathbb{R} \longrightarrow R = \infty$

⑥ - $\sin u = \sum_{n=0}^{\infty} \frac{(-1)^n u^{2n+1}}{(2n+1)!} \longrightarrow I = \mathbb{R} \longrightarrow R = \infty$

⑦ - $\sinh u = \sum_{n=0}^{\infty} \frac{u^{2n+1}}{(2n+1)!} \longrightarrow I = \mathbb{R} \longrightarrow R = \infty$

⑧ - $\cos u = \sum_{n=0}^{\infty} \frac{(-1)^n u^{2n}}{(2n)!} \longrightarrow I = \mathbb{R} \longrightarrow R = \infty$

⑨ - $\cosh u = \sum_{n=0}^{\infty} \frac{u^{2n}}{(2n)!} \longrightarrow I = \mathbb{R} \longrightarrow R = \infty$

Recall:

① - $\sin x \cos x = \frac{1}{2} \sin(2x)$

② - $\cos^2 x = \frac{1}{2} (1 + \cos(2x))$

③ - $\sin^2 x = \frac{1}{2} (1 - \cos(2x))$

④ - $\cosh^2 x = \frac{1}{2} (\cosh(2x) + 1)$

⑤ - $\sinh^2 x = \frac{1}{2} (\sinh(2x) - 1)$

⑥ - $\sinh x \cosh x = \frac{1}{2} \sinh(2x)$

⑦ - $a^u = e^{u \ln a}$

[I] $f(x) = \frac{x}{4+x^2}$

[examples]

$\frac{1}{1+u} = \sum_{n=0}^{\infty} (-1)^n u^n, |u| < 1$

$\frac{1}{4+x^2} = \frac{1}{4(1+\frac{x^2}{4})} = \sum_{n=0}^{\infty} (-1)^n \left(\frac{x^2}{4}\right)^n, \left|\frac{x^2}{4}\right| < 1$

$\frac{1}{4+x^2} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{4^{n+1}}, |x^2| < 4$

$\frac{(x)^1}{4+x^2} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{4^{n+1}}$

$|x| < 2$

$I = (-2, 2)$
 $R = \frac{2+2}{2} = \frac{4}{2} = 2$

على المشهور

السؤال

[4] $\rightarrow f(x) = \frac{1}{(2+3x)^2}$ نشتق مرة واحدة

$$\frac{1}{1+u} = \sum_{n=0}^{\infty} (-1)^n u^n \text{ و } |u| < 1 \rightarrow \frac{1}{2+3x} = \frac{1}{2(1+\frac{3x}{2})} = \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{3x}{2}\right)^n \text{ و } \left|\frac{3x}{2}\right| < 1$$

$$\left[\frac{1}{2(1+\frac{3x}{2})} = \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{3}{2}\right)^n x^n \right] \rightarrow$$

Diff. w.r.t x both sides:

$$\frac{d}{dx} \left(\frac{1}{x} \right) = \boxed{-\frac{1}{x^2}}$$

$$\frac{\boxed{-1}}{(2+3x)^2} \cdot \boxed{3} = \frac{1}{2} \sum_{n=1}^{\infty} (-1)^n \left(\frac{3}{2}\right)^n n x^{n-1}$$

$-\frac{1}{3}$ نضرب

$$\frac{1}{(2+3x)^2} = -\frac{1}{6} \sum_{n=1}^{\infty} (-1)^n \left(\frac{3}{2}\right)^n n x^{n-1}$$

$$-2 < 3x < 2$$

$$-\frac{2}{3} < x < \frac{2}{3}$$

$$I = \left(-\frac{2}{3}, \frac{2}{3} \right)$$

$$R = \frac{b+a}{2} = \frac{1}{\frac{2}{3}} = \boxed{\frac{3}{2}}$$

عوضه بـ $n=0$ فالجواب
اليحيى تساوي صفر
فالحل هنا اذا لا يتغير